Airway

Airway Management Adult and Pediatric

Airway-

AW- 1 Revised March 3, 2023

Most pre-hospital airway emergencies can be managed with proper BVM technique. If possible utilize 2 person BVM techniques.

An adult is defined in this protocol as a patient ≥ 12 years of age or longer than the length based tape.

Trauma:

Utilize in line cervical motion restriction during intubation, I-gel or BVM use.

Intubation:

Consider intubation when the airway is not protected from aspiration.

Position the patient in the sniffing position ensuring the ear canal is in a straight line with the suprasternal notch. In the obese or pregnant patient, elevate the torso with blankets, towels, etc., to achieve this goal.

Pre-oxygenate using 100% oxygen with BVM and 15 Lpm nasal cannula placed in mouth or nares. Leave cannula oxygen in place during intubation attempts. Pre-oxygenate prior to subsequent attempts.

Nasotracheal intubation requires patient have spontaneous respirations. This procedure is contra-indicated in anatomically disrupted or distorted airways, severe facial trauma, or basal skull fracture. Not to be used in children less than 8.

Continuous wave form capnography is mandatory for the initial confirmation and monitoring of all patients with an ETT or BIAD.

Utilize length based tape to determine ETT size on pediatric patient.

Once a patient has been moved with an ETT or I-gel in place, reconfirm placement.

The bougie or videoscope if available may be utilized to facilitate intubation of the adult. Cricoid pressure or BURP maneuver may assist difficult intubations.

The ETT is the preferred airway of choice. An I-gel may be considered if intubation is unsuccessful or considered to be difficult.

An intubation attempt is defined as passing the laryngoscope blade or ETT past the teeth or inserted into the nasal passage. An intubation attempt should be limited to 30 seconds.

After endotracheal intubation, ventilate with a tidal volume just enough to see chest rise in both the adult and pediatric patient.

Proper assessment and documentation of endotracheal intubation (for both the adult and pediatric patient) requires the medic to:

1. Visualize the tube passing between the vocal cords;
2. Ensure no gastric sounds on auscultation;
3. Ensure good bilateral breath sounds while ventilating through the ETT.
4. Observe the chest rising and falling during ventilation,
5. Confirm placement with waveform capnography.
6. Note size of tube and placement at the lips.
7. Note number of attempts.

Intubation Continued:

Reconfirm placement after any patient movement. Confirm placement with receiving MD prior to moving to receiving facility stretcher. Document this in the run report. Mandatory MD signature is required.

Use DOPE to troubleshoot the I-gel or ETT concerns.

Ventilator:

At the discretion of the attending paramedic, the ventilator can be initiated for patients with little or no respiratory effort once successfully intubated or had an I-gel placed. All procedures are to be performed in accordance with in accordance with Frankfort Fire Department SOP 600.15. Tidal Volume for both adult and pediatric patients will be 6 ml/Kg. The ventilator rate should be the same as the BVM rate.

Begin Initial Medical Care:

When using the BVM in adult and pediatric patients, apply 5 cm PEEP by adjusting the PEEP valve. Do not use in infants.

Basic Airway:

Utilize the head tilt-chin lift or jaw thrust to open the airway.
Suction as needed and limit suctioning to 10 seconds per attempt. Oxygenate between attempts.
Place the OP or NP airway as needed if not contraindicated. Utilize an oral airway when providing BVM.
Utilize manual SMR when indicated.
Continuous pulse oximetry should be utilized in all patients with inadequate respiratory function.

Obstructed airway:

Follow AHA guidelines for obstructed airway on the pediatric and adult patient.

Inadequate Breathing:

Begin BVM: Adult/Adolescent: 14-16 bpm Child: 25 bpm Toddler/Infant < 1 year of age: 30 bpm
Neonate: 40 bpm
Ventilate until visible chest rise.

Cardiac Arrest:

If a paramedic is not on scene at the time CPR is initiated with the BVM, the EMT may insert the I-gel.
Choose proper size I-gel. Insert in accordance with manufacturer's instructions.
Confirm placement with auscultation for bilateral breath sounds. Observe equal rise and fall of the chest.
Secure in accordance with Frankfort Fire and EMS procedure.
BVM with I gel in place at 8-10 bpm regardless of age.

If a paramedic is on scene the EMT may place the I-gel **ONLY** upon paramedic approval.

BVM in cardiac arrest with ETT in place:

Ventilate at a rate of 8-10 bpm regardless of age.

Tracheostomy Care:

In the event a patient has an existing tracheostomy and is in need of oxygen therapy, it may be supplied utilizing a T-connector device or trach cuff. Humidify the oxygen if at all possible.
If a patient with an existing tracheostomy is in need of ventilatory support, utilize a bag mask attached to the tracheostomy tube.
The EMT may suction the tracheostomy tube as needed. Insert the whistletip suction catheter 2 inches in the adult and 1 inch into the child tracheostomy, apply suction as you withdraw the catheter.

ALS Airway Care:

Gastric Distention: If gastric distention or vomiting becomes an issue during ventilation with a BVM or remains concerning following intubation, the paramedic may pass a nasogastric tube once and confirmation affirmed utilizing injection of air only.

Inadequate BLS Airway

The medic may intubate or utilize the I-gel in the pediatric or adult patient as needed. Pre-oxygenate as above. Secure tube.
Medic may pass a 12FR NG tube through the I-gel suction port. Confirm NG tube placement with air.
If the adult I-gel airway becomes inadequate due to vomiting or the need for high airway pressures exists, the medic may intubate through the I-gel, removing it, and confirming ETT placement

The medic may nasally intubate as needed.
Apply two or three sprays of **NeoSynephrine** spray in each nostril to vasoconstrict nasal tissue. Lubricate a nasopharyngeal airway with **2% Xylocaine gel** and insert it into the larger nostril if time allows prior to the intubation. A second one may be placed in the remaining nostril. You may further lubricate the ET tube with the 2% lidocaine gel prior to insertion.

Pre-oxygenate the patient as directed.

A BAAM device may be used to enhance passage of the tube by allowing for monitoring of the patient's breathing.

When efforts to oxygenate, ventilate, and Intubate fail; proceed to cricothyrotomy

Obstructed Airway:

If basic airway management fails to open the airway, attempt direct visualization with the laryngoscope and attempt removal of the object using Magill forceps along with suctioning. Attempt BVM. If all these procedures prove unsuccessful, proceed to cricothyrotomy

Pediatric Cricothyrotomy:

a. Locate the cricothyroid membrane and prep the site.
b. Utilize the transtracheal catheter attached to a 10 ml syringe.
c. Direct the device into the midline of the cricothyroid membrane at the angle of 90° while applying negative pressure to the syringe during insertion.
d. Entrance of air into the syringe indicates that the needle is in the trachea. Dip the needle to a 45 degree angle. The catheter is advanced over the needle; the needle and syringe are withdrawn; and the stop-cock attachment is connected to the catheter. An oxygen source with a driving pressure of at least 50 psi is attached, and the three-way stop-cock is operated to ventilate the patient. An inspiratory to expiratory ratio of 1 to 4 may be adequate. Should the chest remain inflated during the period of exhalation, a longer expiratory time, up to 8 seconds, if necessary, may be used.

d. Document the procedure as outlined.

If the patient has a replacement tracheostomy, confirm proper trach size and replace it with the obturator in place. Lubricate with water soluble gel or normal saline. Securely hold the flanges and remove the obturator, ventilate to confirm placement and secure the ties. If the tube cannot be inserted easily, remove it and attempt to pass a smaller sized tracheostomy tube if available or an ETT. Always transport this patient for further evaluation of the placement.

ALS Care Continued:

Obstructed Tracheostomy:

a. Open airway. Make sure the obturator has been removed.
b. Suction the trach utilizing sterile instilled saline. Catheter size is 2 times the internal diameter of the trach. Adult may use 12FR or 14FR.
c. Attempt ventilation. If unable to ventilate and if the tracheostomy tube has an inner cannula, remove it.
d. If this fails to clear the airway, remove the tracheostomy using an outward and downward motion or attempt to place an ETT into the existing tracheostomy tube.
   AT MD DISCRETION ONLY Stitches may be clipped to facilitate removal of a tracheostomy tube.
   If the trach is cuffed, deflate the cuff prior to removing.
e. Replace with new trach or ETT as above.

I gel Insertion Competency
Frankfort Fire and EMS
Franklin County Fire Department

Name: _____ Date: _____

Step	Met	No Met
Don Standard precautions! **EMT use only in cardiac arrest**		
1. Choose correct size I gel based on Kg weight. Use length based tape to determine Kg on pediatric patient.		
2. Open Pack Adult: Place lubricant into smaller plastic bowl. Pediatric: Place lubricant onto indentation of cradle.		
3. Lubricate the back, sides, and front of the I gel. Check to ensure no bolus off lubricant is in the bowl of the I gel obstructing inlet.		
4. Avoid touching the cuff		
5. Place patient in sniffing position unless c-spine is of concern. Then keep patient in neutral position. Pull down on the chin of the patient. Do not introduce fingers into mouth.		
6. Grasp I gel and introduce it with tip toward back of hard palate.		
7. Glide I gel downward and backwards with gentle pressure until a definite resistance is met. Early resistance may be overcome by applying a jaw thrust or twisting the I gel slightly. Do not use excessive force.		
8. The teeth should be resting on the bite block in close proximity to the horizontal line marker.		
9. Adult: Secure the I gel with the support strap. Do not apply tightly over the neck vessels. Pediatric: Secure with twill tape or tape.		
10. Check adequate placement by auscultating breath sounds and looking at rise and fall of the chest.		
11. Ensure the oxygen port cap is ON.		
12. PARAMEDIC ONLY: At your discretion for adult only Place lubricant over the suction port. Insert a 12 fr NG tube and initially prime it. Insert tube down lumen and check placement by auscultating the epigastrium while placing air into the NG tube. Gurgling sounds ensure proper placement. The NG tube may be left open to air or intermittent suction may be applied.		
13. Document size of I gel, and validation of proper placement.		
14. PARAMEDIC ONLY: If there is concern the I gel is not securing the airway, you may attempt intubation through the I gel device.		

Evaluator: _____

Surgical Cricothyrotomy
Frankfort Fire and EMS

NOTE: There are many different techniques for surgical cricothyrotomy; the procedure described here is one example based on commonly available equipment, but modifications based on available equipment may be necessary

Indication:

Inability to oxygenate/ ventilate
AND
Inability to adequately manage the airway via other means

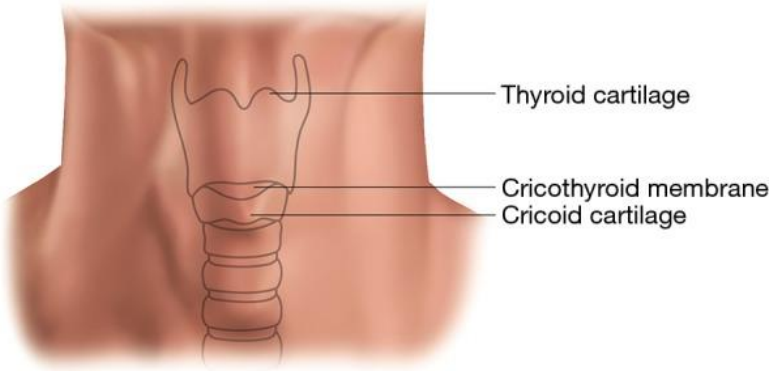
Contraindications:

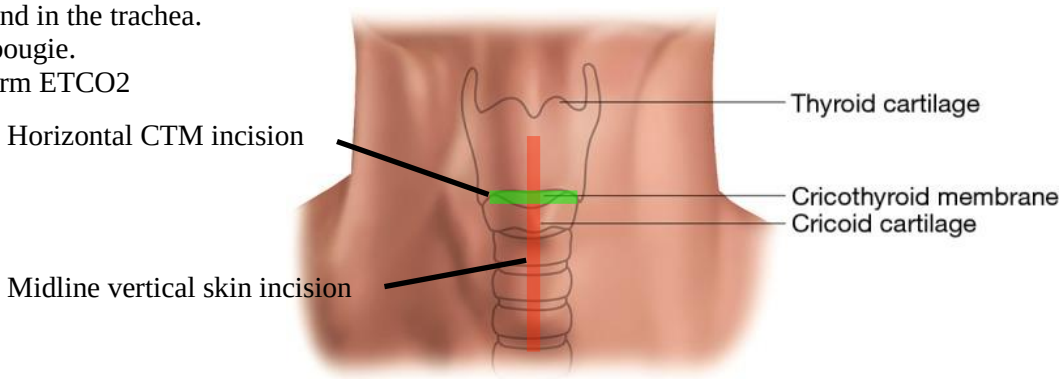
Age < 10 years (needle/ transtracheal jet ventilation preferred, or use of pediatric percutaneous device)

Preparation

-Wear eye protection
-Attempt to oxygenate/ ventilate patient via 2 person BVM or using SGA
-Prepare equipment:
- Scalpel (#10 or #11)
- Kelly forceps
- Bougie introducer
- ET Tube (cuffed, 5.5 or 6.0) or
- 10 ml syringe to inflate balloon

-Position patient supine with head
-Identify landmarks by palpation
- Midline neck
- Thyroid cartilage
- Cricothyroid space
- Sternal notch



Procedure

-If time allows, apply skin disinfectant to proposed site of incision
-Grasp and stabilize the thyroid cartilage with the non-dominant hand.
-Use index finger to palpate the thyroid prominence and the cricothyroiod space
-Make a generous vertical skin incision from the thyroid prominence to below the cricothyroid space.
-Use the Kelly clamp and your finger to dissect through soft tissue until you are able to palpate the cricothyroid membrane. Spread clamp jaws along the direction of the incision.
-Use the scalpel to pierce the cricothyoid membrane and make a horizontal incision through the membrane
-Use your finger to palpate the hole in the cricothyroid membrane, and then insert bougie through the incision and into the trachea
-Pass the ET tube over the bougie introducer and into the trachea just until balloon is through the cricothyroid membrane and in the trachea.
-Inflate the cuff and remove the bougie.
-Confirm placement with waveform ETCO2